



# DEXSeq output tables summary

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|----------------------------------|----------------------------------------------------------------------------------------------------------------|
| ■ Date of Update                 | @August 15, 2025                                                                                               |
| ■ Updated by                     |  Celeste Nobrega              |
| 📁 What Project are you updating? |  <a href="#">cmoore.3REAP</a> |
| ✓ What task are you updating?    | <a href="#">DEXseq</a>                                                                                         |

## APA direction calculation

### 1. Identify each gene's distal PAS

- Using genomic coordinates and strand from the **GRanges**, determine on a per gene basis which PAS is farthest in the 3' direction. On the **+** strand this is the largest coordinate. On the **-** strand this is the smallest coordinate. That PAS is tagged as "distal." All other PAS within the same gene are "proximal."
- This is a custom step, not done by DEXSeq, however, it is conceptually similar to how APA tools separate distal and proximal sites using annotated PAS and strand.

### 2. Quantify condition effects at each PAS with DEXSeq

- DEXSeq fits a negative binomial generalized linear model to counts with a design that tests whether usage of each exon-like feature (here, PAS) changes between conditions.
- It reports a test statistic, p value, adjusted p value, and an effect size column log2fold change that compares Treatment to Control after normalizing for gene-level expression.
- In short, log2 fold change asks whether a given PAS is used more or less often in Treatment than in Control, controlling for the overall gene.

### 3. Translate the sign of the effect into APA direction

- For a **distal** PAS:
  - positive log2 fold change → distal site is used more in Treatment → 3' UTR is **lengthened**
  - negative log2 fold change → distal site is used less → 3' UTR is **shortened**
- For a **proximal** PAS, the logic flips:
  - positive log2 fold change at a proximal site implies **shortening**
  - negative implies **lengthening**
- If the needed inputs are missing, it's labeled **Ambiguous**.

This maps DEXSeq's differential usage signal onto the biological idea of 3' UTR length change. DEXSeq itself provides the differential usage statistics; the distal vs proximal call and the final "Lengthened" or "Shortened" label come from the added logic above.

## Output columns

- **groupID**
  - The gene identifier used by DEXSeq to group features that belong to the same gene. DEXSeq tests usage differences within these groups.
- **featureID**
  - The PAS identifier within the gene. This corresponds to a specific polyadenylation site used as an "exon-like" feature in DEXSeq.
- **gene**
  - A human-readable gene label paired to groupID
- **feature**
  - A human-readable feature label paired to featureID
- **Intron\_exon\_location**
  - Annotation from PAS table describing the site's location category

- many possible values: 3' most exon, 5' most exon, intergenic, Internal exon, Intron, Single exon
- **PAS\_type**
  - Annotation from PAS table indicating PAS class
  - many possible values: 3'UTR(F), 3'UTR(L), 3'UTR(M), 3'UTR(S), 5UTR, CDS, intergenic, Intron, LncRNA(FANTOM5), PAS type, Pseudogene
- **APA\_direction**
  - Biological interpretation of the effect at this PAS in Treatment vs Control: **Lengthened, Shortened, or Ambiguous**, based on distal vs proximal status and the sign of the log2 fold change (see section above).
- **exonBaseMean**
  - Average normalized count for the feature across samples, on the scale DEXSeq uses for modeling. Think “typical coverage” for this PAS.
- **dispersion**
  - The dispersion parameter of the negative binomial model for this feature. Higher values mean more variability between replicates after accounting for mean expression.
- **stat**
  - The test statistic from DEXSeq’s generalized linear model comparison that evaluates differential usage between conditions at this feature.
- **pvalue**
  - Raw p value for the differential usage test at this feature. Lower values give stronger evidence that usage differs between conditions.
- **padj**
  - P value adjusted for multiple testing across features, typically using Benjamini–Hochberg FDR, as implemented by DEXSeq. Use this to decide significance while controlling false discoveries.
- **log2fold\_Treatment\_Control**

- Effect size for Treatment vs Control from estimateExonFoldChanges. Positive means higher relative usage of this PAS in Treatment. Negative means lower. This is the column APA direction uses.
- **countData.[condition\_rep]**
  - Normalized counts used by DEXSeq per sample for this feature.
- **genomicData.seqnames, genomicData.start, genomicData.end, genomicData.width, genomicData.strand**
  - Genomic coordinates and strand for the feature, carried through from the ranges and used to make the distal vs proximal call.